Practice Assignment 11 - Not Graded

This assignment will not be graded and is only for practice.

1 point

Let $f(x, y)$f(x, y) be defined in a neighbourhood of (a, b)(a, b) and $D(x, y) = f_{xx} f_{yy} - f_{xy}^2$ D(x, y) = fxxfyy − fxy2. Suppose (a, b)(a, b) is a critical point of ff. Which of the following option(s) is(are) true?

- [ ] If $D(a, b) > 0$D(a, b) > 0 and $f_{xx}(a, b) > 0$fxx(a, b) > 0, then (a, b)(a, b) is a point of local minimum.
- [ ] If $D(a, b) < 0$D(a, b) < 0, then (a, b)(a, b) is a point of local maximum.
- [ ] If $D(a, b) = 0$D(a, b) = 0, then (a, b)(a, b) may be a point of minimum or maximum or neither.
- [ ] If $D(a, b) < 0$D(a, b) < 0, then (a, b)(a, b) is a saddle point.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If $D(a, b) > 0$D(a, b) > 0 and $f_{xx}(a, b) > 0$fxx(a, b) > 0, then (a, b)(a, b) is a point of local minimum.
If $D(a, b) = 0$D(a, b) = 0, then (a, b)(a, b) may be a point of minimum or maximum or neither.
If $D(a, b) < 0$D(a, b) < 0, then (a, b)(a, b) is a saddle point.

1 point

Choose the correct statement(s) about $f(x, y) = x^4 + y^4$ f(x, y) = x4 + y4, where $D(x, y) = f_{xx} f_{yy} - f_{xy}^2$ D(x, y) = fxxfyy − fxy2.

- [ ] $(0, 0)$(0, 0) is the only critical point.
- [ ] $D(0, 0) > 0$D(0, 0) > 0 and $f_{xx}(0, 0) < 0$fxx(0, 0) < 0 and so $(0, 0)$(0, 0) is a point of local maximum.
- [ ] $D(0, 0) = 0$D(0, 0) = 0.
- [ ] $(0, 0)$(0, 0) is a point of local minimum.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(0, 0)$(0, 0) is the only critical point.
$D(0, 0) = 0$D(0, 0) = 0.
$(0, 0)$(0, 0) is a point of local minimum.

1 point

Let $f(x, y) = (9x^2 - 1)(1 + 4y)$f(x, y) = (9x2 − 1)(1 + 4y) be defined on the box $-2 \le x \le 3$−2 ≤ x ≤ 3 and $-1 \le y \le 4$−1 ≤ y ≤ 4. Then which of the following options is (are) true?

- ○ $(0, 1)$(0, 1) is a point of local minimum.
- ○ $(3, 4)$(3, 4) is a point of absolute maximum.
- ○ -105−105 is the absolute minimum.

○
$(-2, -1)$(−2,−1) is a point of absolute minimum.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(3, 4)$(3,4) is a point of absolute maximum.

1 point

Suppose $z = f(x) + yg(x)$z = f(x) + yg(x) for some polynomial functions ff and gg. Which of the following options is true?

○
$z_{yy} = 0$zyy = 0

○
z_{yy} zyy is a non-constant function of xx.

○
z_{yy} zyy is a non-constant function of yy.

○
$z_{yy} = 1$zyy = 1

No, the answer is incorrect.

Score: 0

Accepted Answers:

$z_{yy} = 0$zyy = 0

1 point

Define a function

$$f(x, y)\text{f(x, y)} = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2} & if\ (x, y) \neq (0, 0) \\ 0 & if\ (x, y) = (0, 0) \end{cases}\{\text{x2+y2xy(x2−y2)0if (x, y)} =$$

$(0, 0)$if (x, y) = (0, 0)

Choose the set of correct options.

☐
$f_{xy}(0, 0) = -1$fxy(0, 0) = −1

☐
$f_{yx}(0, 0) = 1$fyx(0, 0) = 1

☐
$f_{xy}(0, 0) = f_{yx}(0, 0)$fxy(0, 0) = fyx(0, 0)

☐
$f_x(0, 0) = f_y(0, 0)$fx(0, 0) = fy(0, 0)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f_{xy}(0, 0) = -1$fxy(0, 0) = −1

$f_{yx}(0, 0) = 1$fyx(0, 0) = 1

$f_x(0, 0) = f_y(0, 0)$fx(0, 0) = fy(0, 0)

1 point

Consider a function $f(x, y, z) = x^3 y + yz$f(x, y, z) = x3y + yz. If AA is the the Hessian matrix of $f(x, y, z)$f(x, y, z) at (1,1,0), then which of the following option(s) is(are) true?

☐
$det(A) = 0$det(A) = 0

☐

$Rank(A)$ = 2Rank(A) = 2

☐ $Nullity(A)$ = 1Nullity(A) = 1

☐ $det(A) = -6$det(A) = −6

☐ $Rank(A)$ = 3Rank(A) = 3

No, the answer is incorrect.

Score: 0

Accepted Answers:

$det(A) = -6$det(A) = −6

$Rank(A)$ = 3Rank(A) = 3

1 point

Let $f: R^6 \rightarrow R$f : R6 → R be a polynomial function. Which of the following options is true?

○ Hessian matrix of the function is a symmetric matrix of order 6.

○ All possible mixed partial derivatives of second order have the same value at a point of R^6 R6

○ Hessian matrix of the function is a symmetric matrix of order 5.

○ Hessian matrix of the function is a symmetric matrix of order 4.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Hessian matrix of the function is a symmetric matrix of order 6.

Let S be a rectangular box. Let x, yx, y and zz be the length of the sides of the box and volume of the box $V = xyz$V = xyz. If dd is the length of the diagonal of the rectangular box, then $d = \sqrt{x^2 + y^2 + z^2}$d = x2 + y2 + z2 . Use this information to answer the questions 8 and 9.

1 point

If the diagonal is of length 2 units and the volume of the box is the maximum possible, then which of the following option(s) is (are) true?

☐ SS is a cube.

☐ All sides are of different length.

☐ $x = \frac{2}{\sqrt{3}}$x = 3 2

☐ $y = \sqrt{\frac{1}{3}}$y = 31

☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

SS is a cube.

$x = \frac{2}{\sqrt{3}}$x = 3 2

√
2
1 point
If the length of the diagonal of the box is 3 units, then which of the followings is the maximum possible volume of the box?

○ √3 3 √ cube units.

○ 3√33 3 √ cube units.

○ 1 cube units.

○ 33 cube units

No, the answer is incorrect.
Score: 0
Accepted Answers:
3√33 3 √ cube units.

1 point
Consider the following function

$$f(x, y, z) = x^2 y^2 + x z^2 \; \mathrm{f(x, y, z) = x2y2 + xz2}$$

Which of the following option(s) is (are) true?

☐ $(0, y, 0)$, where $y \in R$ (0, y, 0), where y ∈ R are critical points.

☐ $(0, 0, z)$, where $z \in R$ (0, 0, z), where z ∈ R are critical points.

☐ (1, y, 0) where $y \in R$ (1, y, 0) where y ∈ R are critical points.

☐ $(0, y, 1)$, where $y \in R$ (0, y, 1), where y ∈ R are critical points.

☐ $(x, 0, 0)$, where $x \in R$ (x, 0, 0), where x ∈ R are critical points.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(0, y, 0)$, where $y \in R$ (0, y, 0), where y ∈ R are critical points.
$(x, 0, 0)$, where $x \in R$ (x, 0, 0), where x ∈ R are critical points.

Check Answers
Your score is: 0/10